

TASK-1

```
C:\WINDOWS\system32\CMD X + v
Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\raftaar>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4828:400::] with 32 bytes of data:
Reply from 2001:4860:4828:400::: time=175ms
Reply from 2001:4860:4828:400::: time=182ms
Reply from 2001:4860:4828:400::: time=204ms
Reply from 2001:4860:4828:400::: time=107ms

Ping statistics for 2001:4860:4828:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 107ms, Maximum = 204ms, Average = 167ms

C:\Users\raftaar>
```

TASK-2

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\raftaar>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2600:140f:1e00::1737:f4f3]
over a maximum of 30 hops:

  1    3 ms    2 ms    2 ms  2401:4900:ab43:e9a4:8ce8:dfd7:8c98:5da
  2    *      *      *      Request timed out.
  3   126 ms   101 ms  102 ms  fc00:0:2:1f3::1
  4    85 ms    46 ms   36 ms  2404:a800:7a00:102::6
  5    23 ms    53 ms   30 ms  2404:a800:7a00:102::5
  6   141 ms   101 ms  100 ms  2404:a800:88::2
  7   115 ms    65 ms   52 ms  ae8.gpx-bom2.netarch.akamai.com [2600:140f:1e00:ffff::248]
  8    47 ms    53 ms   38 ms  g2600-140f-1e00-0000-0000-0000-1737-f4f3.deploy.static.akamaitechnologies.com [2600:140f:1e00::1737:f4f3]

Trace complete.

C:\Users\raftaar>
```

TASK-3

```
C:\WINDOWS\system32\cmd.  ×  +  v

Microsoft Windows [Version 10.0.26200.8973]
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C:\Users\raftaar> nslookup www.zomato.com
Server:  UnKnown
Address:  fe80::7414:d0ff:fe59:8264

Non-authoritative answer:
Name:     e70929.dsca.akamaiedge.net
Addresses: 2600:140f:4::17da:f9b3
           2600:140f:4::17da:f99a
           23.55.244.160
           23.55.244.186
           23.55.244.185
Aliases:  www.zomato.com
          zomato.edgekey.net

C:\Users\raftaar>|
```

TASK-4

```
C:\>ping 192.168.1.25

Pinging 192.168.1.25 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

TASK-5

Short Report: Using Network Diagnostic Tools

During my network tests, I used **ping**, **tracert**, and **nslookup** to diagnose different types of network problems. For **(a) a website not loading**, **ping** is useful because it checks whether the destination server is reachable; for example, I tested `www.youtube.com` and observed the response time. For **(b) a slow connection**, **tracert** is the best tool because it shows every router (hop) between my computer and the destination; I used `tracert www.flipkart.com` to view the route and identify where delays might occur. For **(c) suspected DNS issues**, **nslookup** is the most useful because it checks whether a domain name is correctly translated into an IP address; I used `nslookup www.zomato.com` to view the resolved IP address. Together, these tools help identify whether a problem is caused by connectivity, routing, or DNS resolution.

